EPPS6323 Knowledge Mining

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EPPS6323

Assignment 3: Prompt exercise

Objective: Design prompts to conduct a structured systematic literature review on data mining and machine learning using multiple AI models (e.g. ChatGPT, Copilot, and Grok 3) and analyze each model's contribution and refining prompts for academic precision and depth.

Exercise Steps:

1. **Step 1: Initial Prompt Creation** ○ Task: Write a baseline prompt to request a structured systematic literature review on data mining and machine learning applications.
   - Example Prompt: "Conduct a 2,000-word structured systematic literature review on the applications of data mining and machine learning in real-world domains. Include a methodology section, synthesize key findings, identify trends and gaps, and propose one testable hypothesis. Use an academic tone and emulate systematic review standards."
   - Students submit this prompt to ChatGPT, Copilot, and Grok 3, collecting the raw outputs.

   Ans:
   Give a Systematic Literature Review on the application of data mining and machine learning in real world scenarios, involving methodology, key findings, trends and gaps as well as a testable hypothesis in about 2,000 words with an academic tone.

2. **Step 2: Analyze Model Responses** ○ Task: Assess each model's output based on:

- **Structure**: Did it include a methodology section and follow a systematic review format?

  ChatGPT provided a well-structured methodology section, offering detailed information on data sources, search strategy and study selection criteria. It also used a structured and bullet-pointed format for applications, making it clearer and easy to read.

  Grok, while slightly less structured, focused more on research questions and provided a strong trends and gaps analysis, which is an essential part of a systematic review.

However, Copilot lacked depth in methodology, provided fewer details on applications, and had a more generalist approach, making it the weakest in terms of systematic review standards**.**

Overall, ChatGPT excelled in structuring the review, Grok was strong in trends and hypothesis testing, and Copilot was the least methodologically rigorous.

- **Synthesis**: Were key findings from data mining and machine learning applications well-summarized?

  In terms of **synthesizing key findings** from **data mining and machine learning applications**, **ChatGPT and Grok** performed better than **Copilot**,.

  ChatGPT provided a **structured and well-bulletined synthesis**, making it **organized and easy to follow**. It effectively categorized **key findings across five application areas** (Healthcare, Finance, Education, Cybersecurity, and Manufacturing).

  Grok, in contrast, presented its synthesis in a **more paragraph-based format**, which was detailed but lacked readability. However Grok captured **important findings well and connected them to trends** in machine learning.

  Copilot, on the other hand, had the **least detailed synthesis** of key findings. While it covered the same five application areas, its analysis was **less comprehensive and more generalist**, making it weaker in terms of depth and clarity.

  Overall, ChatGPT provided the best-structured synthesis with clear bullet points.
  Grok offered a detailed but less structured synthesis, and Copilot had the weakest and most general synthesis of key findings.

- **Trends and Gaps**: Did it identify meaningful trends and research gaps?

  In analyzing **trends and research gaps**, **Grok** provided the **most detailed insights**, effectively identifying **meaningful trends** and **gaps in the literature** compared to

  Grok:. Grok's trends section offered relatively deeper analysis of emerging advancements. Also, its **research gaps analysis was more informative**,

  ChatGPT and Copilot had **vaguer discussions on trends and research gaps**. While ChatGPT provided **some structured points**, its explanation was **not as in-depth** as Grok's. Copilot had the **least detailed** trends and gaps section, making it the weakest in this aspect.

- **Hypothesis**: Was the proposed hypothesis testable and relevant?

  In terms of **hypothesis formulation**, **Grok** performed the best by providing a **testable and relevant hypothesis** along with an **example** making Grok's approach more **practical and applicable** giving relatively a better hypothesis and how it could be empirically tested.

  ChatGPT and Copilot, proposed more **general suggestions** for testing the hypothesis that didn't include an example . Copilot, in comparison, had the **least detailed approach**, making its hypothesis formulation the concise and with less info among all the three.

- **References**: Are the citations accurate (check using Google Scholar or Semantic Scholar) ○ Students document strengths and weaknesses. For example, ChatGPT might provide broad coverage, Copilot might excel at concise structuring, and I (Grok 3) might offer a unique perspective on emerging trends.

Ans: Regarding References

ChatGPT: Had papers cited from different sources MIT press, nature, Journals from Researchgate, research papers from wide platforms even outside the popular trusted sources.

Grok: Gave a sample crafted APA style references at first but when asked specifically for cited papers gave a list of articles and papers published from science direct and springer, nature

Copilot: references were focused more towards appplications from blog post, article, a book chapter rather than research papers.

Overall, Comparatively Grok cited more trustworthy platforms, Chatgpt stressed on citing more research papers and Copilot relied on articles and blogposts.

3. **Step 3: Refine the Prompt** ○ Task: Revise the prompt to address deficiencies in each model's response, creating three tailored prompts—one for ChatGPT, one for Copilot, and one for Grok 3. ○ Example Refined Prompt for Grok 3: "Imagine you're a data scientist conducting a 2,000-word systematic literature review on how data mining and machine learning are applied in domains like healthcare, finance, and education. Outline a clear methodology,

synthesize key findings with fresh insights, highlight trends and gaps, and propose one bold, testable hypothesis.

Maintain a rigorous academic tone."

- Students test these refined prompts and compare the improved outputs.

Ans:

Give a Systematic Literature Review on the application of data mining and machine learning in real world scenarios, involving methodology, key findings, trends and gaps as well as a testable hypothesis in about 2,000 words with an academic tone but this time do make sure to elaborate more on the testable hypothesis also take care to explain technical terms.

Hypothesis elaboration

Led to a more structured hypothesis section including justification, experimental setup, evaluation metrics and expected outcomes**.**
Improved **clarity on how the hypothesis could be tested** instead of just stating it.

Technical term Explanation

Increased the clarity of **complex machine learning and data mining concepts**.
Ensured definitions for terms like **support vector machines (SVMs), deep learning, anomaly detection, AutoML, and Explainable AI (XAI)**.
Made the content more accessible for readers unfamiliar with the subject.

Overall these refinements deepen the review's research utility making it more actionable and comprehensible without sacrificing its systematic structure.

4. **Step 4: Cross-Model Collaboration** ○ Task: Integrate the best elements from each model's output into a final systematic review. Write a new prompt for the student's preferred model (e.g., Grok 3) to synthesize the results. ○ Example Synthesis Prompt: "Using these drafts from three AI models [paste outputs], produce a 2,000-word structured systematic literature review on data mining and machine learning applications. Combine the strongest methodology, findings, trends, gaps, and hypothesis into a cohesive, academically sound document."
    - Students submit their final review and justify their synthesis decisions.

Overall, ChatGPT's structure was retained for clarity and academic flow.
Copilot's expanded industry applications notably retail made it to aplications.
Grok expanded on the experimental setup, adding the idea of using a **clinical trial with human evaluation** to test model trustworthiness

Additionally, There seemed a **broader domain coverage** from social sciences focused by Grok to retail contributed by Copilot.

Collaborative hypothesis focus (Grok's depth and ChatGPT's clarity).
Grok's **detailed experimental design**, proposing tests and models.

Expanded references with Chatgpt's, Grok's and Copilot combined.

5. **Step 5: Reflection** ○ Task: Write a reflection answering:

- How did each model approach the systematic review differently?
Each model had its own approach for the different sections in literature review as mentioned below:

Abstract and Introduction:
ChatGPT and Grok: Both started off with a good Abstract as well as an Introduction.
Grok did take care to explain in short what Data mining is and how it is different from Machine learning in their introduction, whereas none of the others did.
Co-pilot: No Abstract , included a very short introduction alone.

Research strategy:
Grok weighed upon asking more questions than the other two but
ChatGPT and Copilot came up with more info on Data sources and Search strategy.
Also,ChatGPT was seen to have wide and better search platforms than **Copilot.**

Studies from-to:
ChatGPT: studies from 2015-2024 were considered.
Grok: from 2015-2025
Copilot: 2010-2025

Applications:
All three dealt with 5 applications, of all three **Copilot** had the least details.
Both **ChatGPT and Grok** seemed better with Chatgpt having bulletined it while Grok kept it as paragraphs.
Overall **ChatGPT** seemed better at explaining applications as it did tend to have a more narrowed approach rather than being generalist.

Trends and Gaps in liteature
Grok had a pretty good Trends paragraph, whereas **copilot n ChatGPT** had a vague one.
Similarly with gaps in literature as well **Grok** had more details to give than the other two.

Testing out Hypothesis
ChatGPT and Copilot gave a general suggestion with testing out hypothesis whereas **Grok** explained it using an example.

Conclusion
Conclusions of **ChatGPT and Grok** were pretty much the same, **Copilot** had a less convincing one.

- Which prompt refinements yielded the best results for each model?

Ans: The transition from the original prompt to the refined prompt enhanced the systematic literature review by expanding the **testable hypothesis** from a concise to a much more detailed section.
Adding technical **term explanations** such as for "Random Forests" as averaging decision trees or "CNNs" as neural networks with convolutional filters improved ease of understanding.

- What did you learn about leveraging AI for structured academic reviews?

Ans: Using AI to aid with writing an academic literature review gives a **logical approach** of key sections including **methodology, key findings, trends, gaps and hypothesis formulation.**

AI can be used for summarizing certain ideas and analysis done by the researcher in a clear, concise format helping to write better what they want to convey in a structured academic tone without losing details

AI can **narrow down vast amounts of research**, **identify common themes** and **highlight critical insights** from various sources, though all sources may not be trustworthy, filtering through it and finding the right one though a bit hectic when correctly done can help prevent finding and going directly through papers from the web in most cases.

However, **AI-generated content requires human oversight** to verify **accuracy, credibility and nuanced interpretation** of research findings.